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Real-time, Robust and Adaptive Universal Adversarial Attacks Against Speaker Recognition Systems

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Abstract
Voice user interface (VUI) has become increasingly popular in recent years. Speaker recognition system, as one of the most common VUIs, has emerged as an important technique to facilitate security-required applications and services. In this paper, we propose to design, for the first time, a real-time, robust, and adaptive universal adversarial attack against the state-of-the-art deep neural network (DNN) based speaker recognition systems in the white-box scenario. By developing an audio-agnostic universal perturbation, we can make the DNN-based speaker recognition systems to misidentify the speaker as the adversary-desired target label, with using a single perturbation that can be applied on arbitrary enrolled speaker's voice. In addition, we improve the robustness of our attack by modeling the sound distortions caused by the physical over-the-air propagation through estimating room impulse response (RIR). Moreover, we propose to adaptively adjust the magnitude of perturbations according to each individual utterance via spectral gating. This can further improve the imperceptibility of the adversarial perturbations with minor increase of attack generation time. Experiments on a public dataset of 109 English speakers demonstrate the effectiveness and robustness of the proposed attack. Our attack method achieves average 90% attack success rate on both X-vector and d-vector speaker recognition systems. Meanwhile, our method achieves 100× speedup on attack launching time, as compared to the conventional non-universal attacks.

Keywords Speaker recognition systems · Adversarial examples · Universal adversarial attack

1 Introduction

In the past decades, the way human interact with machines has evolved from traditional graphical user interface (GUI) to more interactive and personalized voice user interface (VUI). The popularity of VUIs has exploded with their integration into various applications such as smartphone, smart vehicles, and standalone smart speakers. In particular, speaker recognition system, as one of the VUI that identifies a person from characteristics of voices, has been seamlessly integrated and used in various security-enhanced applications and services, such as remote voice authentication to prevent fraud in financial services, voice-matched voice assistants that can only respond to the owner's voice, and even suspects identification and criminals detection [6, 15].

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Deep network networks (DNNs), with its superiority over conventional machine learning models (e.g., universal background model-Gaussian mixture model) [21, 25], have been adopted as the computation core of the modern speaker recognition systems. However, recent studies have shown that DNN models are vulnerable to adversarial attack in various fields, e.g., computer vision [40] and natural language processing [8, 10]. Inspired by these phenomena, some recent works have begun to investigate the vulnerability of DNN in audio domain. In [19], Kreuk et al. generates adversarial examples against an end-to-end speaker verification model, which is essentially a two-class speaker recognition system. A more recent work [11] successfully launches the adversarial attack against a more complex multi-class speaker recognition model. Despite these recent reports, the vulnerability of speaker recognition systems against adversarial attack in practical setting is still not fully understood because of three reasons. First, the existing audio-domain attacks [11, 19] are *individual attack* (i.e., non-universal), which means they need to generate different perturbations for each voice input. Since each perturbations requires considerable

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generation time, such voice-specific generation scheme cause real-time attack become impossible. Second, though two very recent works [29, 42] proposes untargeted universal audio perturbations, they only consider a digital scenario where the audio adversarial example is directly fed into the DNN models without suffering any distortion (e.g., reverberations occurred during physical playbacks) or environmental noise. Hence their attack scenarios are overly idealized and unrealistic in real worlds. Third, the existing audio-domain universal adversarial perturbation is directly applied to audio clips without being further optimized. Such absence of utilizing the input information results the current universal audio adversarial perturbation suffers large magnitude, and thereby being easily to be detected.

In order to better understand the vulnerability of DNN-based speaker recognition system against the adversarial attack, in this paper we propose a *real-time*, *robust* and *adaptive universal* targeted adversarial attack against two representative DNN-based multi-class speaker recognition models (X-vector [38] and d-vector [44]). By crafting an audio-agnostic universal perturbation, which can be added into any voice input of any enrolled speaker's, our proposed universal audio-domain adversarial attack can mislead the speaker recognition system to output an adversary-desired speaker label with high attack success rate. More specifically, the technical contributions of our work are summarized as follows:

1. To make the universal perturbation fit different voice inputs with different lengths of duration, we propose to first generate a small fixed length unit universal noise, and then build the desired length of adversarial perturbation on the top of this via repeated playback.
2. Unlike [19] that feeds the adversarial examples to the speaker recognition system directly, we take one step forward – we successfully launch robust adversarial attacks through estimating the distortions incurred by the physical world propagation, thereby making the adversarial examples remain effective after over-the-air playing.
3. Moreover, in order to make the attack to be undetectable, we propose to use adaptive spectral gating to further adjust the perturbation's spectrum energy. Such adjustment is dynamic and fully utilizes the spectral information for each individual input voice, thereby significantly improving the imperceptibility of the attack.

Our experiments on a public dataset of 109 speakers demonstrate the effectiveness and robustness of our proposed attack approach. With a high attack success rate of over 90%, our proposed universal attack only needs around $0.015s$ launching time, which means $100\times$ speedup over conventional attacks. Moreover, after applying the spectral gating to the generated adversarial examples, the voice quality have been significantly improved with remaining high attacking performance. This work is the extension of our prior ICASSP 2020 paper [45].

2 Background

2.1 Adversarial Attack

DNN models are vulnerable to adversarial attack, a type of attack method via feeding adversarial examples as the inputs of DNNs. Specifically, adversarial example is generated by adding the intentionally designed perturbations, namely adversarial perturbation, to the original legitimate examples. With such adversarial perturbations, adversarial examples can cause the DNN models make incorrect classification or predictions [40]. Interestingly, such perturbations are unnoticeable to human's observation, thereby posing severe security challenges for practical deployment of DNN models. Historically, adversarial examples were first investigated in computer vision community [40]. And then their many intriguing properties attract the machine learning community to explore the impact of adversarial examples in other machine learning tasks, such as natural language processing [10], malware detection [17], reinforcement learning [23], etc.

With the different types of adversarial examples, adversarial attacks can be categorized into two classes: untargeted attack and targeted attack. Untargeted attacks are easier to realize since it only requires the model output is incorrect. On the other hand, targeted attack expects the adversarial examples can misguide the model to make a designated prediction or classification. Also, according to the different prior knowledge of adversary, adversarial attacks can be categorized as white-box attack or black-box attack. White-box attacks assume the adversary has full access to the target DNN model, including model architecture, weight parameters and the training data [40]. In contrast, for black-box attack, the adversary only knows the output of the target model without accessing the model architecture and gradient information. For example, Zoo [12] explores a zeroth order optimization to find the adversarial perturbation with the estimated gradient through querying a target model and observing its output.

To date there are many approaches to generate adversarial examples in white-box scenario. Fast gradient sign method (FGSM) [14] is proposed to quickly generate adversarial examples in a one-step manner – it produces adversarial perturbations via the gradient information of the

DNN model. Built on the top of FGSM, iterative FGSM with clipping [20] generates stronger adversarial examples, which, however, would cost longer generation time. Carlini et al. [9] proposes to find the small perturbations using iteration-based optimization. Given the difficulty of high non-linearlity of DNNs, the objective function is reformulated for easy optimization. Although this method can achieve very strong attack performance, it suffers time-consuming perturbation generation.

2.2 Voice Controllable Systems

Voice Controllable systems refer to systems that administrated by voice speech commands spoken by users in a natural language, such as English and Mandarin. Thanks to the growth of computational power, smart and mobile devices are currently equipped with voice controllable systems, e.g., Amazon Alexa [4], Google Assistant [16], Apple Siri [5] and Microsoft Cotana [26]. By taking human speech as input, voice controllable systems allow users use a convenient interaction manner to achieve various tasks. For example, the user can control the light through smart home device by saying "Turn on the light". Moreover, by understanding the human nature language, the voice controllable systems can also ask user's requirements or answer questions to achieve more human-computer interactions.

With this ongoing trend, while much attention is being attracted to improve the capabilities of voice controllable systems, the security property is also being explored. In particular, speaker recognition techniques are seamlessly integrated with voice controllable system to provide personalized services (e.g., online shopping, reading email) associated with a user's account. The injection of voice authentication prevents the systems to execute commands from potentially malicious sources. For example, Amazon Alexa Voice Profile [3] allows to use a user's voiceprint to authorize online purchases. Since in real-world scenarios like corporate office, many voice-controllable system-equipped smart devices, e.g. Amazon Echo and Google Home, are shared by multiple users; those shareable devices can be accessed by stranger or malicious users without being noticed. In this case, malicious attackers can obtain the control surreptitiously over physically-inaccessible systems, thereby putting users' privacy and properties under high risks. Consequently, it is very important and necessary to study the security vulnerability of such voice-controllable systems. In this work, we focus on developing an audio adversarial attack that can effectively spoof speaker recognition schemes in voice controllable systems with a short launching time.

3 Related Work

3.1 Universal Adversarial Attack

Moosavi-Dezfooli et al. first demonstrate the existence of image-domain universal adversarial perturbation, which can be added to arbitrary input image and cause misclassification with a large probability [27]. Brown et al. [7] utilize robust training algorithms to generate adversarial patch that can be placed at arbitrary position of any input to mislead the image classifier to output target class. Several follow-up studies have reported similar observations in other vision tasks such as semantic image segmentation [18], object detection [24], and person detection [41]. In audio domain, recent studies propose a single universal perturbation which can fool automatic speech recognition (ASR) systems and cause errors in transcription [29], or fool speech command classifier to make false prediction [42]. It is noted that these work focus on untargeted attack, in which the adversary cannot freely specify the expected speech transcription or speech command class. Most recently, the study in [2] reports that it is possible to generate targeted audio adversarial perturbation against audio classification models (e.g., environmental sound classification). However, this type of attack only consider a digital scenario, where the crafted audio adversarial examples are directly fed into the DNN models without taking into account real-world distortions, such as absorption and reverberation incurred by over-the-air playback. Moreover, the generated perturbation is directly applied without being optimized for each specific audio, thereby reducing the imperceptibility of the corresponding adversarial audio examples.

3.2 Adversarial Attack on Speech Recognition

Recent studies have successfully produce adversarial examples against automatic speech recognition (ASR) speech-to-text system. For instance, Vaidya et al. [8, 43] generate noise-like adversarial sound to make ASR models output adversary-desired text transcriptions. However, the generated adversarial examples would be perceived by human. To solve this problem, Carlini et al. [10] propose to craft adversarial examples by adding unnoticeable perturbations into original speech. Moreover, CommanderSong [46] is proposed to embed any malicious command into regular songs. Such modified song will be recognized by ASR systems as malicious commands, but still being perceived as common music by human. However, all the above mentioned adversarial attacks are individual attack, which means they need to produce the adversarial example for each individual input audio via solving an optimization problem, and thereby causing very high run-time requirements (e.g., several hours) to generate the adversarial examples per input audio.

3.3 Adversarial Attack on Speaker Recognition

Different from speech recognition systems, speaker recognition systems mainly focus on extracting individual-dependent voice characteristics through embedding methods to identify speakers' identities regardless of their speech content. Recently, encouraged by the superiority of scalable embedding performance [21, 25], a growing trend is to use DNNs in the embedding layers of the speaker recognition models. However, few studies have been conducted to explore the vulnerability of those DNN-based speaker recognition systems. An initial study from Kreuk et al. [19] proposes to build adversarial examples against an end-to-end speaker verification model, which is essentially a two-class speaker recognition system. In this work, the voice distortions incurred by practical over-the-air playback is not considered. To address this issue, a follow-up study [22], based on the FSGM method, successfully launched practical adversarial attack against a multi-class speaker recognition model by measuring the room impulse response (RIR), which encodes the environmental acoustic channel state information (CSI). A more recent work [11] demonstrates the feasibility of launching targeted adversarial attack against commercial speaker recognition systems. However, the above mentioned studies only discuss *individual attack*, which requires a long time to craft different perturbations for each voice input, making it impossible to be launched in practical real-time scenarios. Consequently, the feasibility of crafting adversarial examples that can fool speaker recognition model on arbitrary input with realistic distortions hasn't been fully exploited and investigated yet.

4 Speaker Recognition System & Threat Model

4.1 Target Speaker Recognition Model

X-vector In this work, we will first use the DNN-embedding-based X-vector model [38] as the target speaker recognition system. Recently X-vector models have shown a significant improvement over the standard i-vector models, and therefore receive much attention in several follow-up studies (e.g., [32, 37]). The architecture of X-vector model is shown in Fig. 1a [45]. Here for a input audio, the system first extracts mel-frequency cepstral coefficents (MFCCs) features using a sliding window. The extracted features are then passed to a time-delay neural network (TDNN) [36] that operates on audio frames. The statistics pooling layer takes the output of the final frame-level layer as input, aggregates over the input segment, and computes its mean and standard deviation. Subsequently, hidden layers are used to map the concatenated statistics into

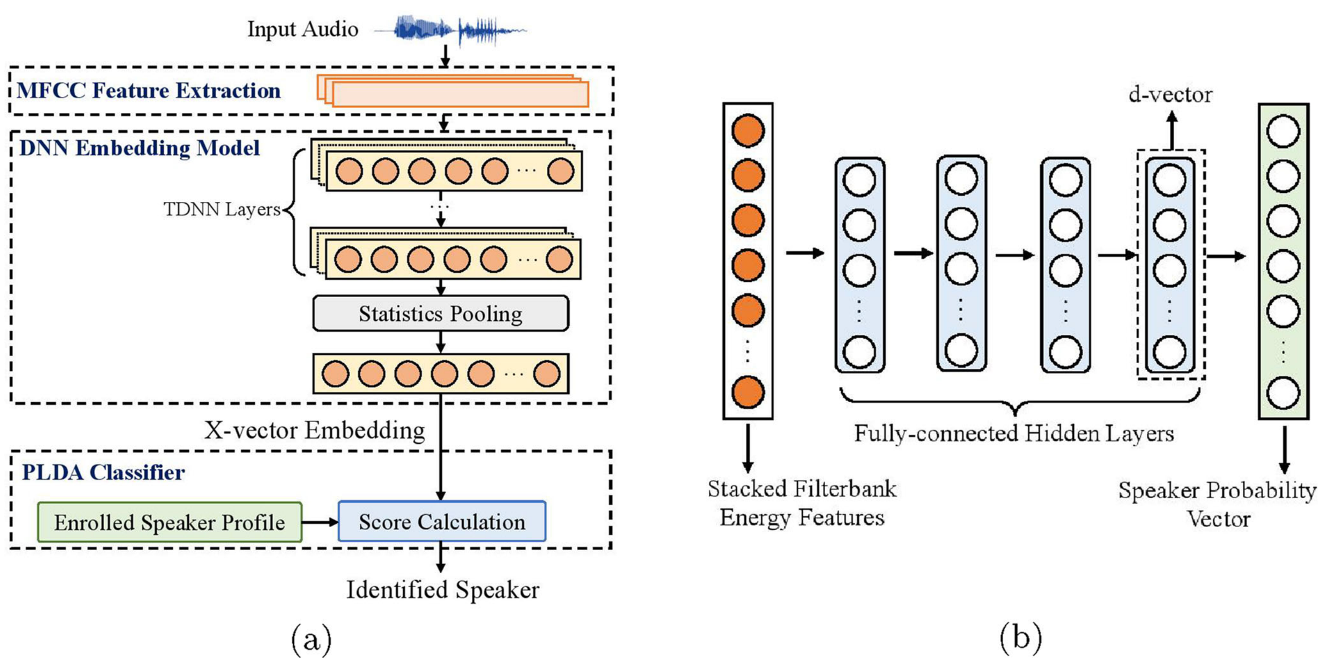

Figure 1 Illustration of the target speaker recognition models

final embeddings. In the recognition phase, the probabilistic linear discriminant analysis (PLDA) module computes the probability of the input audio belonging to each enrolled speaker with the embedding information and identifies the speaker label with the highest calculated score.

d-vector In our experiment we also use d-vector [44] as the second speaker recognition model. d-vector model is proposed to adopt DNN architecture as an feature extractor to automatically learn speaker-specific features during supervised training to replace the conventional i-vector model. Different from x-vector model that consists of TDNN layers, d-vector model use fully-connected layers to operate at frame-level for extracting more compact representation of the speaker acoustic features. Figure 1b illustrates the d-vector structure. Here the entire network consists of 4 fully-connected hidden layers (256 unites per layer) plus rectified linear units [28] (ReLU) serving as the activation function. The last two hidden layers use dropout [39] with dropout rate of 50%. The network receives stacked filterbank energies as the input and then are trained to classify speakers. Once the network is well-optimized, The activation values from the last hidden layer are extracted to form the d-vector.

4.2 Challenges and Threat Model

Challenges In practice, launching an adversarial attack against speaker recognition system in real-world setting is facing a number of challenges:
(1) Timing Challenge. To craft an adversarial perturbation with respect to the speaker's speech, using conventional optimization-based approach is usually very time-consuming, and thereby making many practical attack scenarios, such as playing the adversarial perturbation along with the speaker's voice input, become impossible.
(2) Universality Challenge. To meet the timing requirement in practical setting, a solution is to generate universal adversarial perturbation, which can be repeatedly used for different audio inputs. However, to achieve that a universal mapping from the audio sources to the adversary-desired target must be built, and it is a non-trivial task. Also, such universal perturbation must be general enough and adaptive for different audio inputs with different lengths and spoken by different speakers with various accents.
(3) Robustness Challenge. When we consider audio-domain adversarial attack in the real world, e.g. playing the adversarial examples over the air, the attack performance would be inevitably impacted by the sound distortions incurred by the attenuation and multi-path effects. Thus, the generated adversarial perturbations must be robust enough to remain effective after suffering those real-world distortions.
(4) Imperceptibility Challenge. To maintain high robustness, the strength of the attack should be strong for realizing successful attack. However, too much strong attack will increase the probability of being detected by humans. Therefore, to avoid being noticed the generated adversarial perturbation also must be as imperceptible as possible. A good balance between imperceptibility and attack performance must be achieved.

Threat Model In this paper, white-box attack is used as the threat model. In other words, the adversary has full knowledge of the target speaker recognition model as well as its parameters. Meanwhile, for the attack considering the sound distortions in the room, we assume the adversary also has access to the room's layout, e.g., the size of the room and its reverberation rate. Please note that the placements of the loudspeaker for playing the adversarial example and the microphone of the speaker recognition system are unknown to the adversary.

As shown in Fig. 2, we aim to find a single audio-agnostic universal perturbation that can be applied on arbitrary enrolled speakers' input audio to mislead the speaker recognition system. Additionally, we expect to build a more robust adversarial perturbation that can remain effective during the over-the-air propagation in the simulated environments. The generated universal perturbation should also be further tailored for each individual audio to improve its imperceptibility.

5 Real-time, Universal, Robust and Transferable Audio-domain Adversarial Attacks

5.1 Real-time Attack via Universal Adversarial Perturbations

Most of existing audio adversarial attack studies [10, 19, 31, 46] fool the DNN-based systems in an input-dependent manner – they build different adversary perturbations for different individual input audios. Commonly, they utilize time-consuming multi-step generation methods, such as C&W [9] or iterative FGSM (I-FGSM) [20], to prepare adversarial perturbations. From the perspective of practical deployment, such long generation time limits the feasibility of those methods for real-world deployment. Even worse, for every new voice input such long generation process has to be repeated entirely from scratch due to the input-dependent characteristics of these existing methods, and thereby prohibiting the generation of audio adversarial perturbations against the real-time speaker recognition systems.

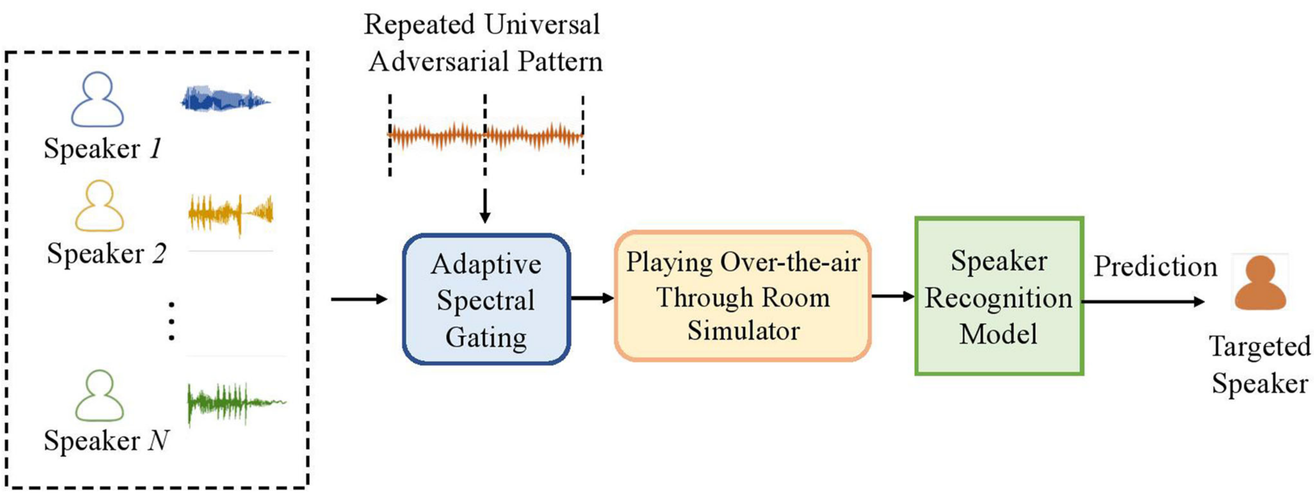

Figure 2 Threat model of the proposed attack

Differently, in this paper we explore how to build a single universal perturbation that can be directly applied to arbitrary speaker's any utterance, making the speaker recognition system output the adversary-desired speaker label. As revealed by its name, this universal perturbation can be applied and re-used on different voice inputs to form different adversarial examples. Such unique input-independent characteristic, by its nature, makes real-time attacks possible.

5.2 Improving Robustness via Room Impulse Response

In physical environments, unlike image data, transmitting acoustic signals over the air will cause distortions of original audio and involving environmental noises. Commonly, acoustic propagation in a room is considered as a linear and time-invariant system. Thus the recorded signal $R(x)$ could be presented as a deterministic function of the played original signal x: $R(x) = x * r$, where r is the estimated room impulse response (RIR), and $*$ denotes the convolution operation. To simulate this play-over-the-air process in the physical world, in this paper we utilize an acoustic room simulator [35] to generate estimated RIR r. Specifically, a simulated room setting is first determined by configuring the 3D shoe-box room size and the the sound absorption coefficient of walls. Next, the placement of the source speaker and target microphone need to be decided and then generate r accordingly. It is important to note that the estimated RIR is closely associated with the locations of the microphone and the loudspeaker in the room.

In order to preserve the effectiveness of the adversarial example after over-the-air propagation, we propose to take the RIR into account in our universal perturbation generating phase. The perturbations are calculated using the gradient information by back-propagation through both the classifier and the RIR transformation. By doing so, optimization associated with the simulated RIR enhances the robustness of the generated adversarial example, and consequently enables over-the-air attack in practice. The detailed implementation is introduced in 5.3.

5.3 Robust Universal Adversarial Perturbation Generation

To clearly present the steps of our perturbation generation, we model the target speaker recognition system, as a function $F(x)$, which takes as input an utterance x and outputs a predicted speaker label. We define $P(x)$ as the function of all DNN layers (including PLDA) to compute the probabilities of classifying x as each of the profiled speakers, and the speaker with highest calculated probability, $F(x) = Argmax(P(x))$, will be recognized. Therefore, to launch a universal targeted adversarial attack, where targeted speaker label is t, we aim to find a perturbation δ that can achieve $F(x+\delta) = Argmax(P(x+\delta)) = t$ for arbitrary x.

To build a universal adversarial perturbation, we need to find a general solution that can make the generated perturbation effective for all the utterances regardless of their speakers, accents, speech content and length. To overcome the issue of varying utterance length, we dynamically construct the universal perturbation δ based on the length of the input utterance as follows:

$$\delta = Crop([\triangle\delta \frown ... \frown \triangle\delta], x), \tag{1}$$

where $\triangle\delta$ is a short-length adversarial perturbation (e.g., its duration is $1s$ in our work), and $[\triangle\delta \frown ... \frown \triangle\delta]$ is a vector constructed by repeating $\triangle\delta$. $Crop(\cdot, \cdot)$ crops the first input to the length of the second input. By using this crop-based strategy, the derived perturbation δ can be applied to the audio input with any length.

To minimize the distortion between the adversarial example and the original voice, δ is clipped to a pre-defined range. The generated adversarial example with the clipped δ is formulated as:

$$x' = x + Clip_{\epsilon}(\delta), \tag{2}$$

where $Clip_{\epsilon}(\delta)$ is the function to perform element-wise clipping of δ. Values of δ outside the interval $[-\epsilon, \epsilon]$ is clipped to the interval edges, and ϵ is our pre-defined attack strength.

Then, we iteratively derive the targeted adversarial example via using the following objective function:

$$Argmax(P(x' * \mathbf{r})) = t, \tag{3}$$

where t is the targeted speaker label, r is the simulated RIR, $*$ denotes the convolution operation, and $x' * r$ represents the estimated adversarial example recorded by the microphone. To make the generated adversarial examples robust in various environmental settings, we estimate multiple RIRs $\mathbf{r}$ in various environments. We randomly select one RIR in $\mathbf{r}$ for each training step when updating the perturbation based on each training utterance. In this way, we aim to make the adversarial perturbation still effective in different RIR settings.

In addition, as directly solving the non-linear constrained non-convex problem is difficult, we iteratively solve the following optimization problem :

$$\textbf{minimize}\ \max(\max\{P(x' * r)_i : i \neq t\} - P(x' * r)_t, -\kappa), \tag{4}$$

where $\{P(x' * r)_i : i \neq t\}$ represents the output probabilities of all speakers except the targeted speaker, while $P(x' * r)_t$ denotes the predicted probability to the targeted speaker. κ is a configurable parameter which represents attack confidence. A high κ value will force the optimizer to find adversarial examples that will be misrecognized as the target class with a high confidence [8, 9]. In our implementation, we set κ to 0. To generate the universal perturbation, we iteratively modify the trainable sequence, $\triangle\delta$, which is used for constructing δ, with the entire training dataset until achieving the desired attack success rate. For each training utterance, if the predicted probability of the targeted class is larger than other classes, the update of the perturbation $\triangle\delta$ is skipped on the next sample. The details of our algorithm are illustrated in Algorithm 1.

Algorithm 1 Robust universal adversarial perturbation generation.

Require: Training dataset $\chi = \{x^{(1)}, ..., , x^{(n)}\}$, Target label t, Speaker recognition model $F(\cdot)$, Perturbation constraint constant ϵ, RIRs for training $\mathbf{r} = \{r_1, ..., r_m\}$, Desired attack success rate γ .
Result: Universal adversarial pattern $\triangle\delta$
Initialize $\triangle\delta$
while Attack Success Rate(χ) $< \gamma$ **do**
  for $x^{(i)} \in \chi$ **do**
    if $Argmax(F(x' * r)) = t$ **then**
      Skip training on $x^{(i)}$
    else
      for number of step **do**
        $\delta = Crop([\triangle\delta \frown ... \frown \triangle\delta], x^{(i)})$;
        $x^{(i)'} = x^{(i)} + Clip_{\epsilon}(\delta)$;
        $r \leftarrow getrandomRIR \in \{r_1, ..., r_m\}$;
        $\boldsymbol{y_{pred}} \leftarrow Argmax(F(x' * r))$;
        Loss $\leftarrow \max(\max\{F(x' * r)_i : i \neq t\} - F(x' * r)_t, -\kappa)$;
        minimize$Loss$to update $\triangle\ \delta$
      end
    end
  end
end

5.4 Enhancing Imperceptibility via Adaptive Spectral Gating

Most existing universal adversarial attacks directly apply a uniform perturbations on the input without any adoption. However, in the case of attacking speaker recognition systems, the hearing quality of adversarial examples may be different for different underlying target utterances, and therefore they can be easily noticed by human. To launch a stealthy attack, we propose to adaptively modify the universal perturbations with respect to each individual utterance in the inference phase to improve the imperceptibility. In particular, the universal adversarial perturbation added to the utterance needs to have a low loudness accordingly. This is very challenging since the universal perturbation is always entangled with the user's utterance in time domain, and hence naively reduce its amplitude will significantly degrade the attack performance. To address this challenge, we propose to utilize the time-frequency relationship of the voice – we make the perturbation less perceptible by reducing its energy in frequency domain, and meanwhile we preserve the audio quality of the utterance.

To be specific, we explore a spectral gating method [34] that finds the spectrum of the perturbations and uses it as a reference to reduce the energy of the adversarial example through frequency-domain thresholding. As shown in Fig. 3, the entire procedure starts by calculating the spectrogram of the perturbation, which cuts the time-series perturbation into overlapped short frames of length L with a step size of S. For each frame, we apply $F - points$ Fast Fourier Transformation (FFT) to derive frequency components and convert the frequency components into dB scale via: $20 \times Log_{10} STFT$, where $STFT$ denotes the spectrogram in amplitude. The threshold for each individual frame is set based on the statistics of the frequency components:

$$Thres_t = m_t + std_t \times n_{thres}, \tag{5}$$

where m_t and std_t are the mean and standard of frame t. n_{thres} is a hyperparameter to control the strength of denoising. By calculating thresholds in the frame level, our algorithm can perform fine-grain perturbation reduction.

Next, we apply the frame-level thresholds on the individual spectrogram of adversarial examples to reduce the loudness of the perturbation. Similar to the generation of perturbation, we calculate $F - points$ spectrogram, $STFT_{adv}$, in dB scale with the same frame length and step size. For frame t, we then compare the energy with $Thres_t$ and create a mask M for the component with energy lower than the thresholds. The mask M is multiplied with a scaling factor p to control the energy of the denoised spectrogram. To avoid abrupt changes in the consecutive spectrogram components, we smooth the M with a Gaussian kernel over time and frequency domain. We then apply the smoothed mask M' on $STFT_{adv}$ through spectral gating:

$$STFT'_{adv} = STFT_{adv} \times (1 - M) + g \times M, \tag{6}$$

where g is the minimum energy of $STFT_{adv}$ in dB. Finally, the algorithm converts $STFT'_{adv}$ from dB scale to amplitude scale and performs inverse short-time Fourier transformation to reconstruct the adversarial example in the time domain. By utilizing the proposed spectral gating method, the perturbation is further tailored according to each individual audio, and therefore making the generated adversarial example more imperceptible to human.

6 Experimental Results

6.1 Experimental Methodology

Dataset We evaluate our proposed attack on an English multi-speaker corpus provided in CSTR voice cloning toolkit (VCTK) [13]. In total, the dataset contains 44217 utterances spoken by 109 speakers with various accents. The dataset is divided into a training and a testing set with a ratio of 4:1.

Baseline Model We implement the X-vector and d-vector systems using TensorFlow [1]. For X-vector system [38], 30-dimensional MFCC features with a frame length of $25ms$ are extracted. A pre-trained X-vector DNN embedding model provided in Kaldi [30] is used in the model. The baseline model achieves a classification accuracy of 92.8% on 8896 testing utterances from 109 speakers. In our implementation, we train the d-vector model to directly classify the speakers in the training set in an end-to-end manner. After training 100 epochs via Adam optimizer with the learning rate of 0.001, d-vector model achieves classification accuracy as 86.45% on test data.

Evaluation Metrics *(1) Attack Success Rate*: The ratio between the number of successful attacks and the total number of attack attempts; *(2) Noise Level*: We quantify the relative noise level of the perturbation δ with respect to the original audio x in decibels (dB): $D(\delta, x) = 20log_{10}(\frac{max(\delta)}{max(x)})$. (3) *PESQ* : We use perceptual evaluation of speech quality (PESQ) [33] score to quantify the speech quality by human perception. By comparing the degraded signal with the provided a reference signal, a score is calculated ranging from -0.5 to 4.5, where a larger score indicates better speech quality.

6.2 Attack Evaluation

Speedup on Attack Time Unlike conventional individual attacks that require to build adversarial perturbation

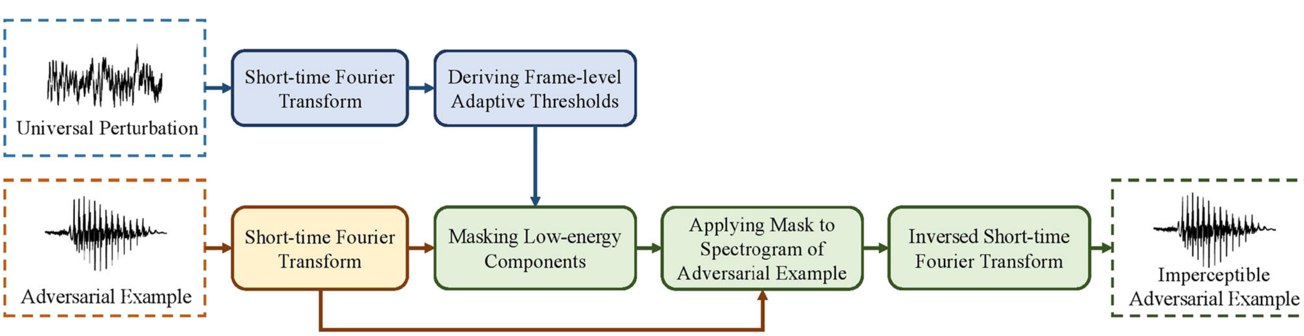

Figure 3 Illustration of applying adaptive spectral gating on an adversarial example

for each individual voice input, our proposed universal attack generates a single perturbation that makes arbitrary speaker's utterances to be identified as the adversary-desired speaker. Therefore, the real-time audio-domain adversarial attack can be realized via simply playing the pre-generated universal perturbation.

We compare the attack launching time of using the conventional individual targeted attack method [10] and our proposed universal attack for a given audio signal. Particularly, the conventional targeted attack requires at least $15s$ to be launched, measured on a Tesla V100 GPU with $32GB$ memory; while our proposed universal method only needs an average of $0.015s$, thereby bringing $100\times$ speedup.

Effectiveness of Universal Targeted Attack To evaluate the effectiveness of our universal targeted attack, we alternatively choose one of the 109 enrolled speakers as the targeted speaker and the rest 108 speakers as victims. In total, we generated 109 universal adversarial perturbations, trying to make the speaker recognition system classify the victims' utterances as the targeted speakers.

As shown in Table 1, by adjusting attack strength ϵ, the noise level of the generated adversarial perturbation ranges from $-18.84dB$ to $-33.96dB$. As discussed in the previous study [10], such noise level is considered to be quasi-imperceptible to humans. For instance, $-33.96dB$ is comparatively the difference between a person talking and the ambient noise in a quiet room. For each ϵ value, the minimum, maximum, and average attack success rate among all attack attempts targeting on 109 speakers are calculated. We observe that when the noise level is $-18.84dB$, a high average attack success rate of 99.95% can be reached on X-vector model. When the noise level decreases to -33.96 dB, the average attack success rate still remains over 80%. Moreover, the attacking performance on d-vector model is demonstrated in Table 1b. The average attack success rates are also comparable to those of the X-vector system. In overall these experimental results show the effectiveness of our proposed universal targeted attack across different DNN-based speaker recognition systems.

Robustness Analysis Using Room Simulator An acoustic room simulator toolkit [35] is used to simulate the audio propagation in a room environment. Specifically, a modeled room with a size of $5m \times 5m \times 3m$ is used, and 120 locations of the loudspeaker and the microphone are chosen randomly in the room for RIR estimation. For the estimated RIRs, 100 locations are used to build the universal, targeted and robust adversarial perturbation, and the rest 20 locations are used for testing.

Table 2 summarizes the results of our practical universal adversarial perturbation. It is seen that the universal adversarial perturbations trained with RIRs still remain effective after the over-the-air propagation. In particular, the practical universal perturbation generated with a noise level of $-18.84dB$ can still achieve an average attack success rate of 90.19% and 90.32% for X-vector and d-vector, respectively. For comparison, we test the adversarial perturbation of the same noise level without RIR in the simulated room environment. In such setting the average attack success rate decreases significantly to 1.33% and 1.87% on our target speaker recognition systems, respectively. This phenomenon shows that our approach can efficiently improve the robustness of the generated adversarial examples.

Imperceptibility via Adaptive Spectral Gating (ASG) In order to further improve the quality of our robust universal perturbation, we utilize spectral gating according to each target input utterance to adjust the magnitude of the perturbations. To demonstrate the effectiveness of the ASG, we visualize and compare the power spectrogram of generated adversarial examples before and after the proposed spectral gating. As shown in Fig. 4b, the high magnitude adversarial perturbations make the acoustic spectrogram noisy as compared to the original audio

Table 1 Results of universal targeted attack on speaker recognition systems

Attack Strength	Noise Level	Min. Attack Success Rate	Max. Attack Success Rate	Avg. Attack Success Rate
(a) X-vector.				
ϵ=0.05	-18.84dB	98.47%	100%	99.95%
ϵ=0.03	-23.27dB	95.31%	99.91%	98.40%
ϵ=0.01	-33.96dB	53.32%	95.48%	83.82%
(b) d-vector.				
ϵ=0.05	-18.84dB	89.01%	99.93%	98.20%
ϵ=0.03	-23.27dB	76.82%	96.90%	93.15%
ϵ=0.01	-33.96dB	59.64%	94.29%	81.77%

Table 2 Results of robust universal targeted attack using acoustic room simulator on speaker recognition systems

	Noise Level	Min. Attack Success Rate	Max. Attack Success Rate	Avg.Attack Success Rate
(a) X-vector.				
Without RIR	-18.84dB	0.70%	3.52%	1.33%
	-18.84dB	74.68%	98.05%	90.19%
With RIR	-23.27dB	66.54%	96.81%	86.17%
	-33.96dB	54.48%	90.83%	78.25%
(b) d-vector.				
Without RIR	-18.84dB	0.33%	4.21%	1.87%
	-18.84dB	76.08%	99.20%	90.32%
With RIR	-23.27dB	58.98%	95.36%	83.29%
	-33.96dB	55.65%	89.13%	77.02%

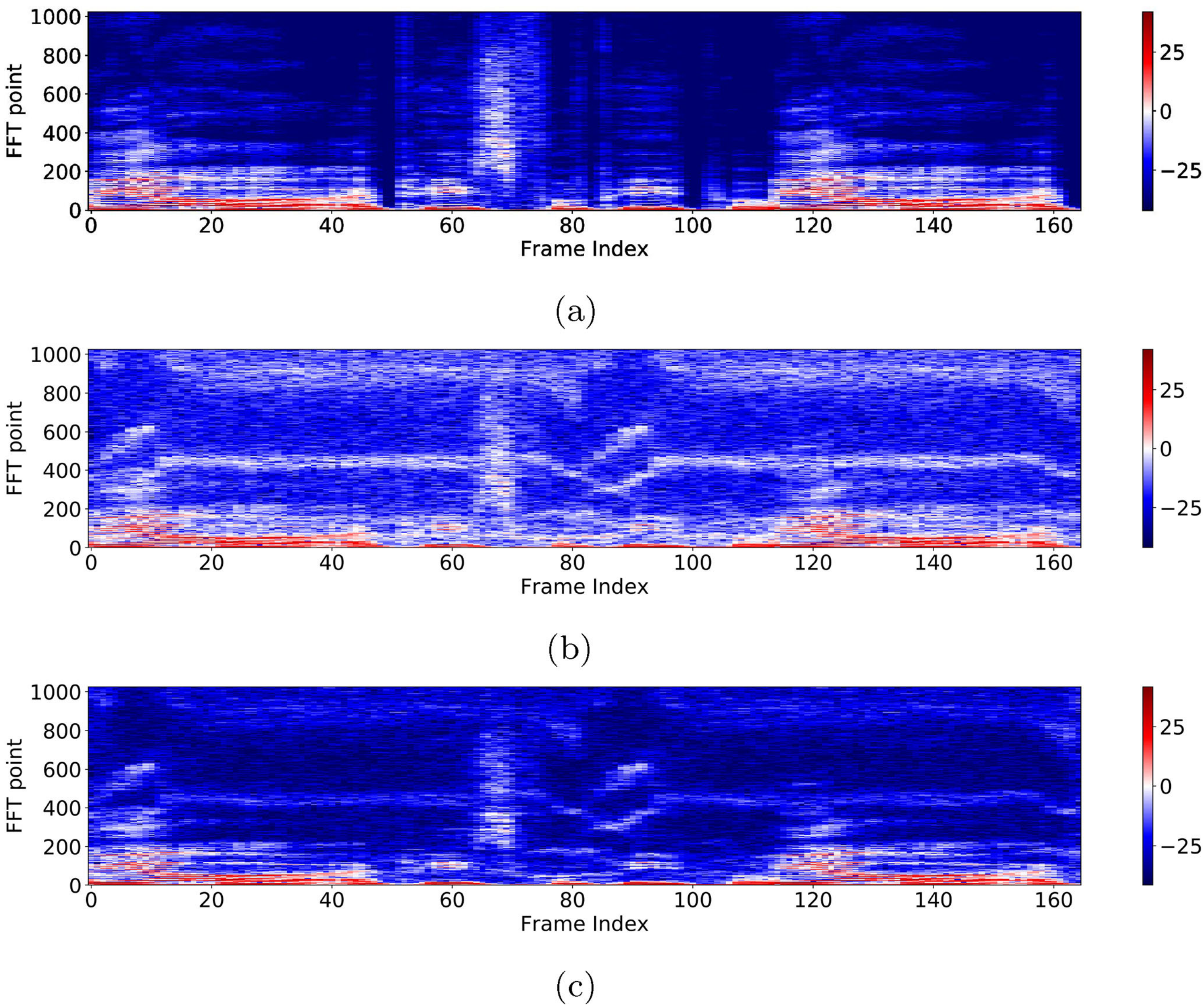

Figure 4 Power spectrogram of the generated adversarial example

Table 3 Results of robust universal targeted attack using adaptive spectral gating (ASG) on speaker recoginition systems

	Avg. Attack Success Rate		Avg. PESQ	
	without ASG	with ASG	without ASG	with ASG
(a) X-vector.				
-18.84dB	90.19%	89.36%	1.21	1.48
-23.27dB	86.17%	83.25%	1.58	1.92
-33.96dB	78.25%	73.45%	2.16	2.45
(b) d-vector.				
-18.84dB	90.32%	89.33%	1.17	1.35
-23.27dB	83.29%	80.33%	1.39	1.64
-33.96dB	77.02%	73.98%	2.12	2.40

spectrogram shown in Fig. 4a. Such perturbations can be easily noticed by human. Figure 4c shows the spectrogram of the same adversarial example after passing our proposed spectral gating (with scaling factor $p = 0.8$). It is seen that the energy of the perturbations is relatively suppressed, and thereby improving the imperceptibility of adversarial perturbations.

Moreover, we also demonstrate the attacking performance and hearing qualify of applying spectral gating mechanism on the adversarial examples in Table 3. By setting the scaling factor $p = 0.5$, the average attack success rate on both X-vector and d-vector are only slightly reduced no more than 4%, with still remaining over 89%, 80% and 73% for the noise level as -18.84dB, -23,27dB and -33.96dB, respectively. The scaling factor p can be delicately determined to balance between the attack success rate and the speech quality. On the other hand, the PESQ score increases, so the speech quality by human perception is still ensured. In addition, though it costs around $0.5s$ to generate the perturbation, our proposed adaptive universal attack still enjoys $30\times$ speedup comparing to the conventional individual targeted attack method [10], thereby demonstrating the feasibility to launch such robust and adaptive universal adversarial attacks in a real-time manner.

7 Conclusion

This paper proposes a real-time, robust and adaptive universal targeted adversarial attack against speaker recognition systems. The proposed attack builds a universal perturbation that can be added into any enrolled speaker's voice input to mislead the system to output any adversary-desired speaker label. The robustness of the adversarial perturbations is also greatly improved by using an acoustic room simulator to estimate the sound distortions associated with the over-the-air propagation. Moreover, spectral gating is proposed to improve the imperceptibility and make the generated universal perturbation adaptive to fit each individual input. Evaluation on a public dataset of 109 speakers shows the effectiveness and robustness of our proposed attack.

In our future work, we plan to explore the following directions: first, besides simulated environments, the attack can be extend to physical environments by replacing the estimated RIRs with physically measured RIRs from real-world environments; second, we plan to explore black-box attacks on commercial speaker recognition systems; and lastly, we will explore extending the proposed targeted universal attack to other intelligent audio systems such as automatic speech recognition.

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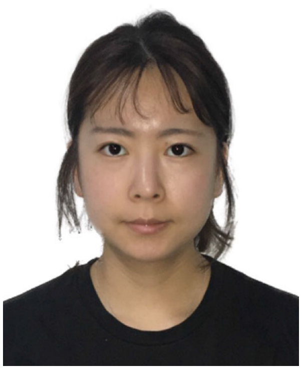

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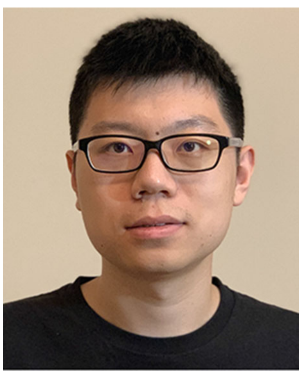

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